ROUGEHALLA / OPTIMIZED PLAY LOOP

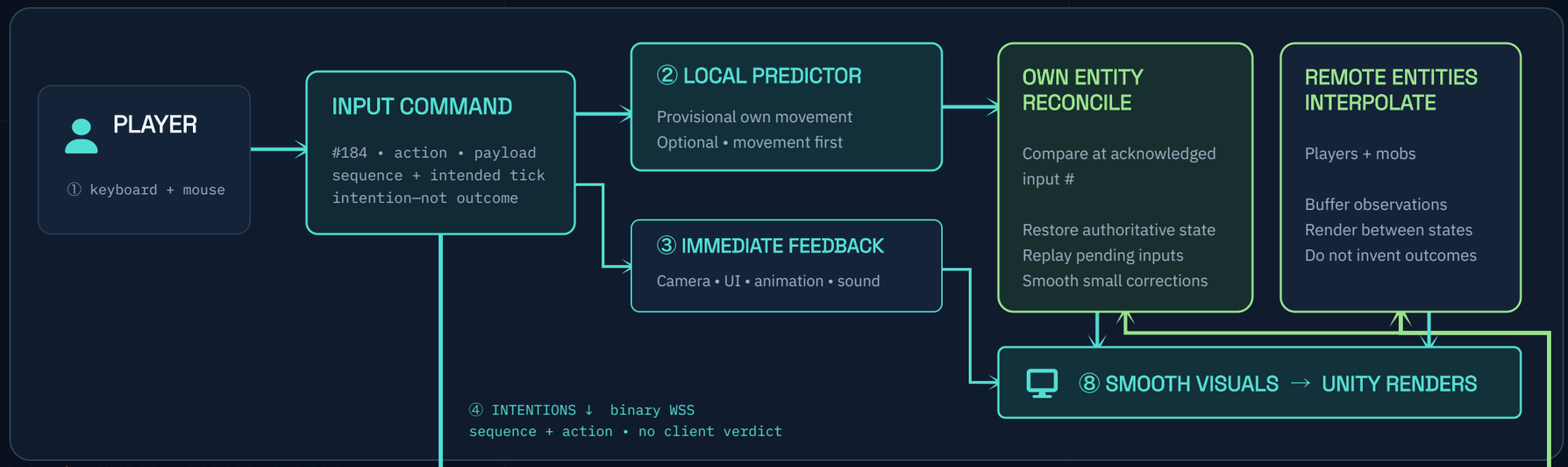
Predict locally. Decide centrally.

Immediate presentation. Authoritative Rust. Reconciled Unity visuals.

TARGET ARCHITECTURE

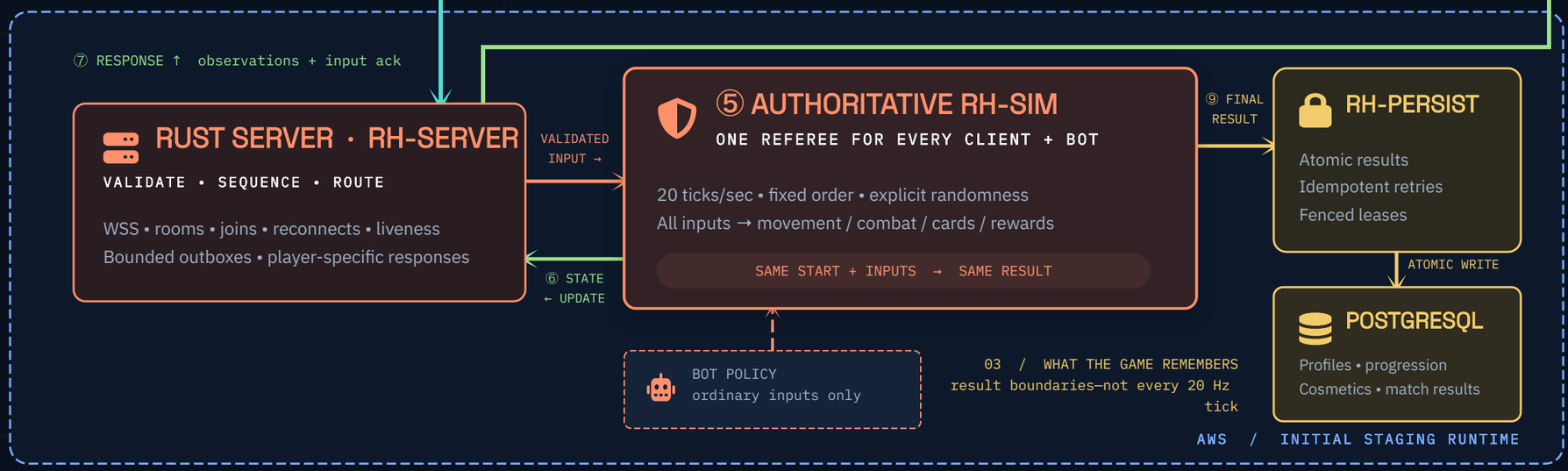
Prediction is a staged enhancement—not a claim that the port or rollout is finished.

01 / PLAYER'S BROWSER · UNITY WEBGL CLIENT



02 / WHAT DECIDES THE TRUTH

TRUST BOUNDARY • PREDICTED ≠ VERIFIED



04 / HOW WE KEEP IT HONEST

evidence + human approval

- **HEADLESS TESTS** — Same simulation, no browser
- **REPLAYS + HASHES** — Same inputs → same result
- **PARITY + BALANCE** — Legacy evidence, declared bands
- **GATES + HUMAN MERGE** — Compile, lint, test, review

RECOMMENDED ROLLOUT: authoritative baseline → immediate feedback → remote interpolation → own-movement prediction + reconciliation. Consider shared Rust/WASM prediction only after replay parity is dependable; test latency, jitter, loss, reconnects and corrections.

THE THREE DISTINCTIONS

- **Predict = provisional**
  Local Rust may forecast responsive own movement. It cannot verify hits, rewards or match outcomes.
- **Decide = authoritative**
  The server combines all inputs, hidden state and randomness to advance the real match.
- **Reconcile = converge**
  Restore at the acknowledged input, replay pending inputs, then smooth small visual corrections.

WHAT MAY APPEAR WHEN

IMMEDIATE	Camera, UI, animation, sound; provisional own movement.
CONFIRMED	Damage, deaths, random cards, loot, quests and match results.
REMOTE	Interpolate approved observations; do not predict their outcomes.

WHAT IS ACTUALLY OPTIMIZED

PERCEIVED INPUT LATENCY

Local feedback appears now. Unity renders above 20 Hz while Rust advances the match at 20 Hz.

DATA + DATABASE WORK

Compact inputs + player-specific observations avoid sending hidden world state for prediction. PostgreSQL writes only at deliberate result boundaries.

PREDICTION DOES NOT REMOVE SERVER WORK

Bandwidth is not automatically reduced. Real load tests still determine match density and hosting capacity.

PREDICT CAREFULLY • VERIFY ALWAYS